

Title : Implement Client Script & UI Policy (Incident)

College Name :

College Code :

University Name : Manonmaniam Sundaranar University, Tirunelveli

Course Name : ServiceNow – System Admin with IT Service
Management & AI Emphasis

Department : B.sc Information Technology

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Implement Client Script & UI Policy (Incident)

Introduction:

Incident records require consistent and accurate data entry for effective triage, routing, and resolution. Relying only on user awareness and manual checks can lead to incomplete, inconsistent, or incorrect information. This project demonstrates how ServiceNow UI Policies and Client Scripts can control field behavior, enforce mandatory values, automate updates, and validate Incident records before submission.

The implementation uses UI Policy Actions together with onChange, onSubmit, and onCellEdit Client Scripts. The configuration dynamically responds to the Impact field, controls the Urgency field, validates Assigned To, and prevents direct State changes through list editing.

Problem Statement

Incident records often require consistent and accurate data entry to ensure effective triage, routing, and resolution. However, relying solely on user awareness and manual checks can lead to incomplete, inconsistent, or incorrect data being submitted. This can impact reporting accuracy, SLA compliance, and overall service quality. To address this challenge, conditional field behavior and validation are enforced directly at the user interface level.

Objective

The objective of the Implement Client Script & UI Policy (Incident) project is to demonstrate how ServiceNow client-side controls can be used to enforce data integrity on Incident records. The implementation dynamically makes fields mandatory, auto-populates values, controls field behavior, and prevents record submission when required conditions are not met.

Skills

Incident Management

UI Policy

UI Policy Actions

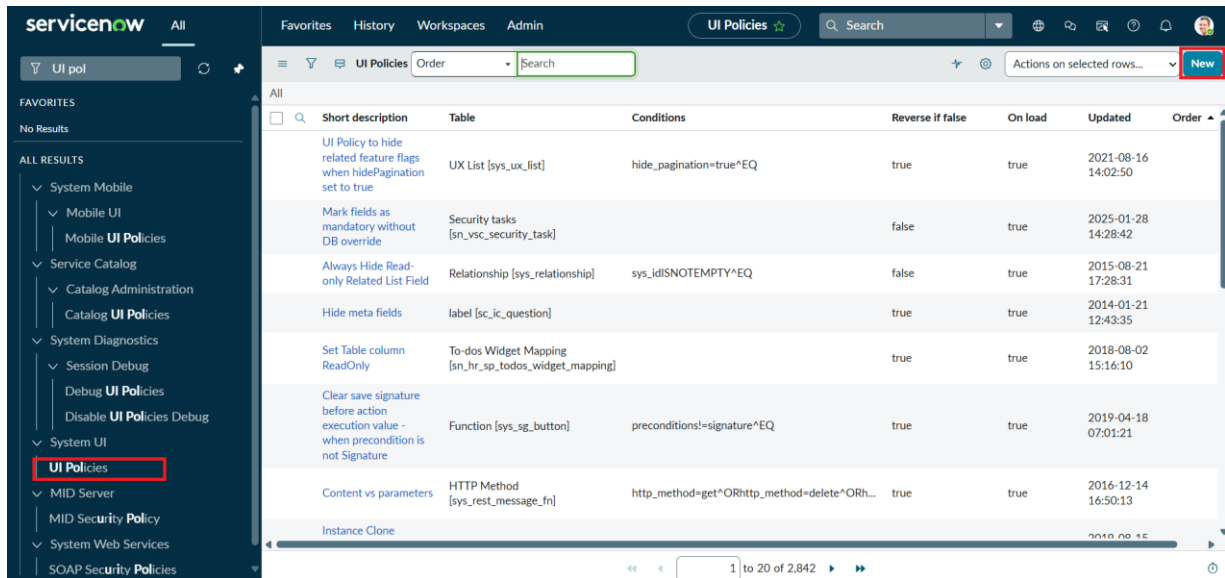
Client Scripts

Form Validation

Milestone 1: Create UI Policy on Incident

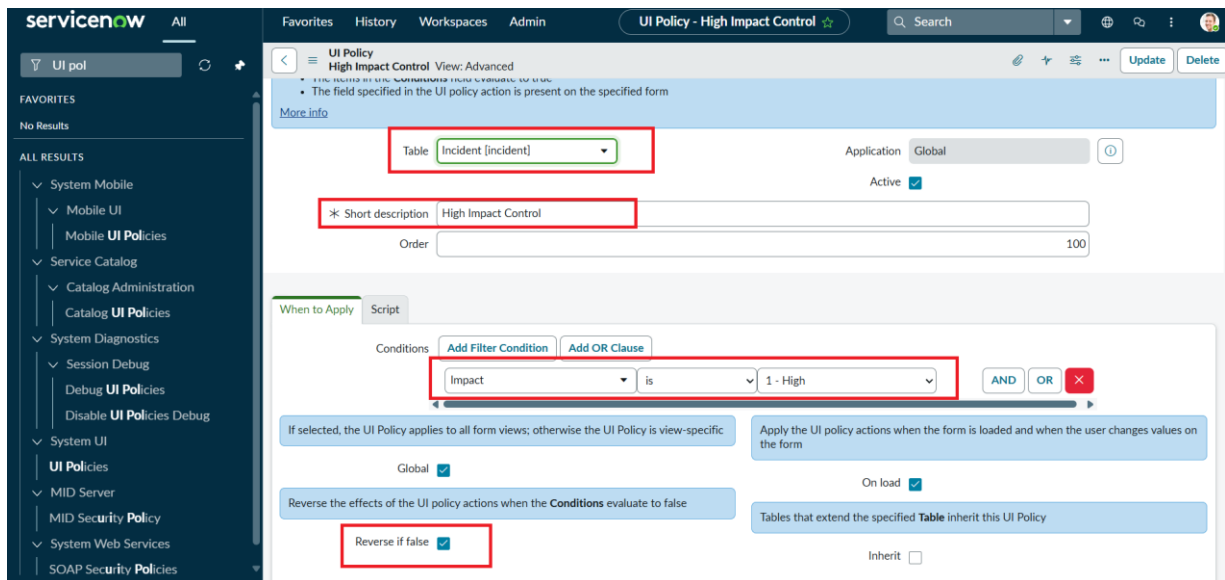
Activity 1: Create UI Policy

1. Log in to the ServiceNow instance with administrative or configuration access.
2. Navigate to System UI → UI Policies.



Screenshot 1: Navigating to UI Policies.

3. Click New to create a new UI Policy record.

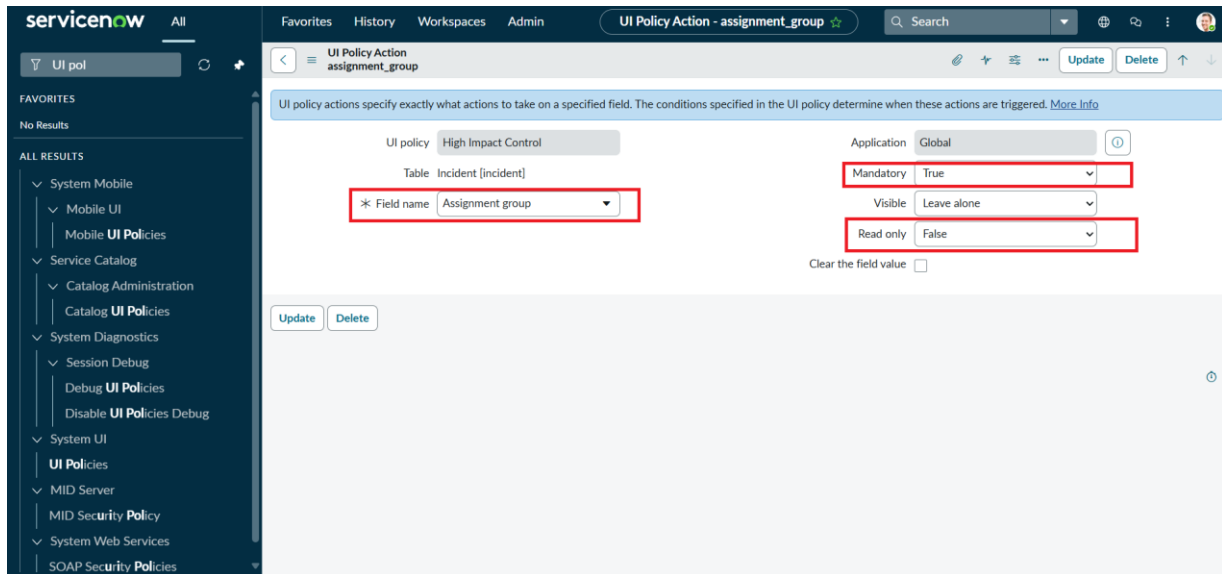


Screenshot 2: Creating the High Impact Control UI Policy.

Fill in the following details:

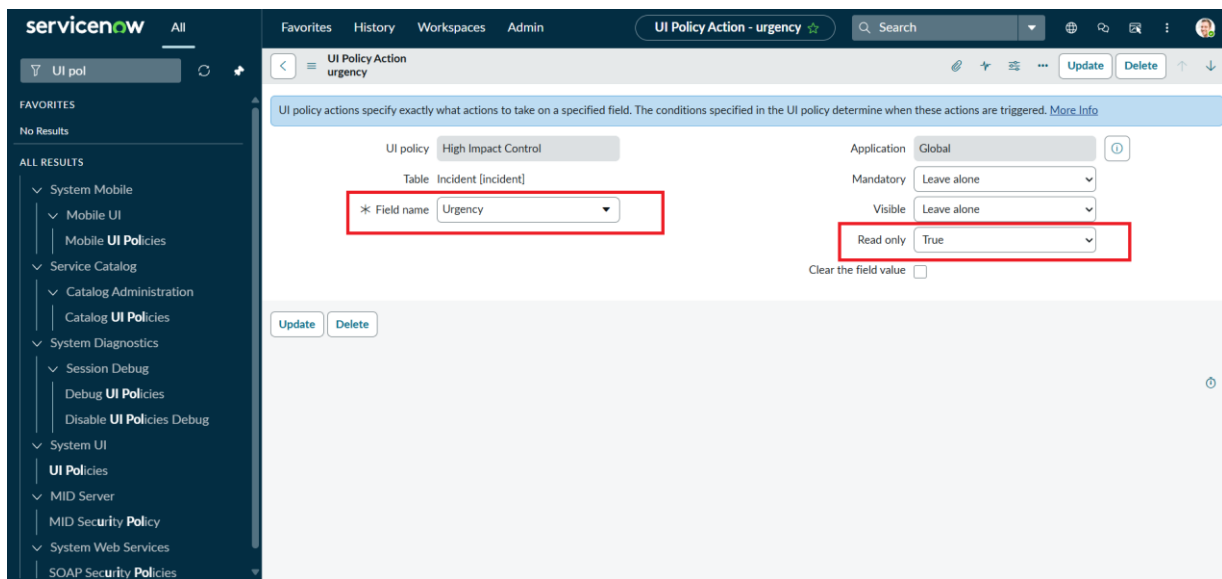
- **Name:** High Impact Control
- **Table:** Incident
- **Active:** true

4. Configure the Conditions section: Field = Impact, Operator = is, Value = 1 – High.



Screenshot 3: Configuring the Impact = 1 – High condition.

5. Configure the UI Policy Action for Assignment group and check Mandatory.



Screenshot 4: Making Assignment group mandatory.

6. Set Reverse if false to true. This reverses the field changes when the condition is no longer true.

7. Review the configuration and click Submit to save the UI Policy.

Milestone 2: Create UI Policy Action – Urgency

Activity 1: Configure the Urgency Field

1. Open the High Impact Control UI Policy and scroll to the UI Policy Actions related list.
2. Click New to create a UI Policy Action.

Name	Active	Table	Application	View	Type	Updated
Maximum Duration Range Check	true	Transaction Quota Rule [sysrule_quota]	Global		onChange	2012-05-30 14:47:01
Create default value	true	Response Template Channel [sn_template_srip_channel]	Templated Snippets		onChange	2020-09-22 17:06:27
Show overlap field based on data labels	true	Widgets [sa_widgets]	Global		onChange	2021-04-23 02:59:51
Set protocol based on auth type	false	HTTP(s) Connection [http_connection]	Global		onChange	2017-06-02 15:49:24
Set default values for OIDC PKCE flow	true	OAuth OIDC Entity [oauth_oidc_entity]	Global		onChange	2020-02-28 00:59:58
Make notification read only	true	Additional Notification Recipients [sys_notification_recipient]	Global		onLoad	2021-02-05 22:42:13
Deprecated - SN - Make serial number man	false	Asset [alm_asset]	Global		onChange	2025-03-06 00:42:54
Validate State/Set Name	true	MID KBs Deployment Request [mid_kbs_deployment]	Global		onChange	2022-10-09 23:18:07
Set Use current record if delete	true	Action Item [sys_sg_write_back_action_item]	Global		onChange	2019-04-14 07:30:02
Warning when no destination type enabled	true	Notification [sys_notification]	Global		onLoad	2021-05-13 04:51:51
Sync Status Change Handler	true	Advanced Portal Navigation [sn_ex_sp_portal_extensible_navigation]	Employee Center		onChange	2024-12-23 03:29:15
Populate, hide/show relation cons table	true	Finding Constraint [promin_finding_def_constraint]	Global		onChange	2022-09-12 07:27:29
Mark type read only	true	System Address Filters [sys_email_system_filter]	Global		onLoad	2020-02-17 02:09:09
Populate Process Definition	true	Activity [sys_pd_activity]	Global		onLoad	2022-08-22 13:59:47
Populate downgrade from license	true	Software Upgrade and Downgrades [mdb_m2m_downgrade_model]	Global		onChange	2014-02-03 16:49:33
Set Activity Definition Variables	true	Workflow UI Policy Action [wf_ui_policy_action]	Global		onChange	2016-02-05 07:22:26
Set file name extn	true	Export Set [sys_export_set]	Global		onLoad	2017-05-23 16:49:28

Screenshot 5: Opening the UI Policy Actions list.

3. Set Field name = Urgency, Read-only = true, and leave Visible unchanged.

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Auto set urgency for high impact
 Table: Incident [incident]
 UI Type: Desktop
 Type: onChange
 Field name: Impact

Application: Global
 Active: ☒
 Inherited: ☐
 Global: ☒

Description:

Messages:

Script

```

1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || newValue == '') {
3     return;
4   }
5
6   if (newValue == '1') { // High Impact
7     g_form.setValue('urgency', '1');
8     g_form.showFieldMsg(
9       'Impact',
10      'High Impact incidents must have High urgency and an assignment group.',
11      'Info'
12    );
13  }
14 }
15

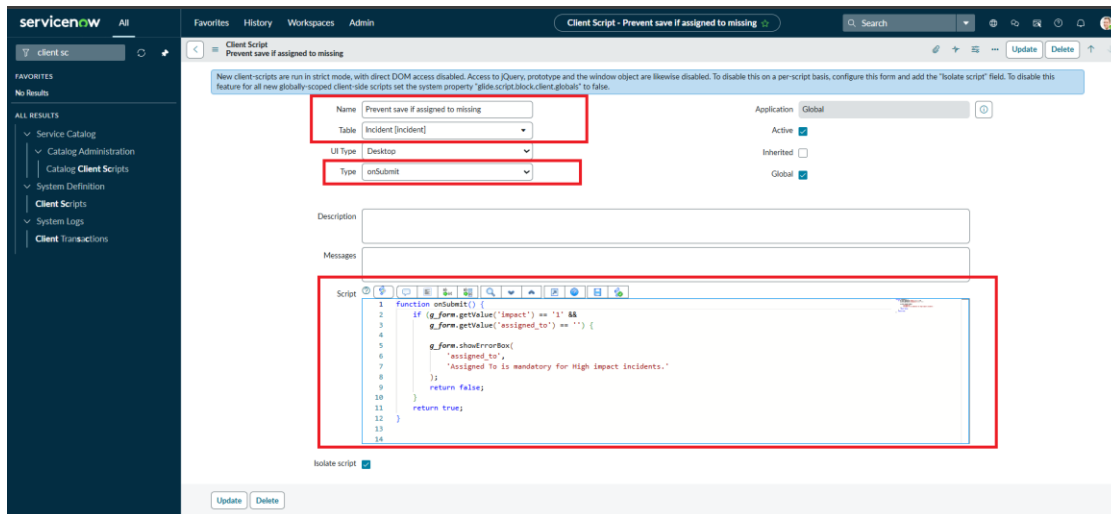
```

Isolate script: ☒

Update Delete

Screenshot 6: Configuring the Urgency UI Policy Action.

4. Click Submit to save the action.



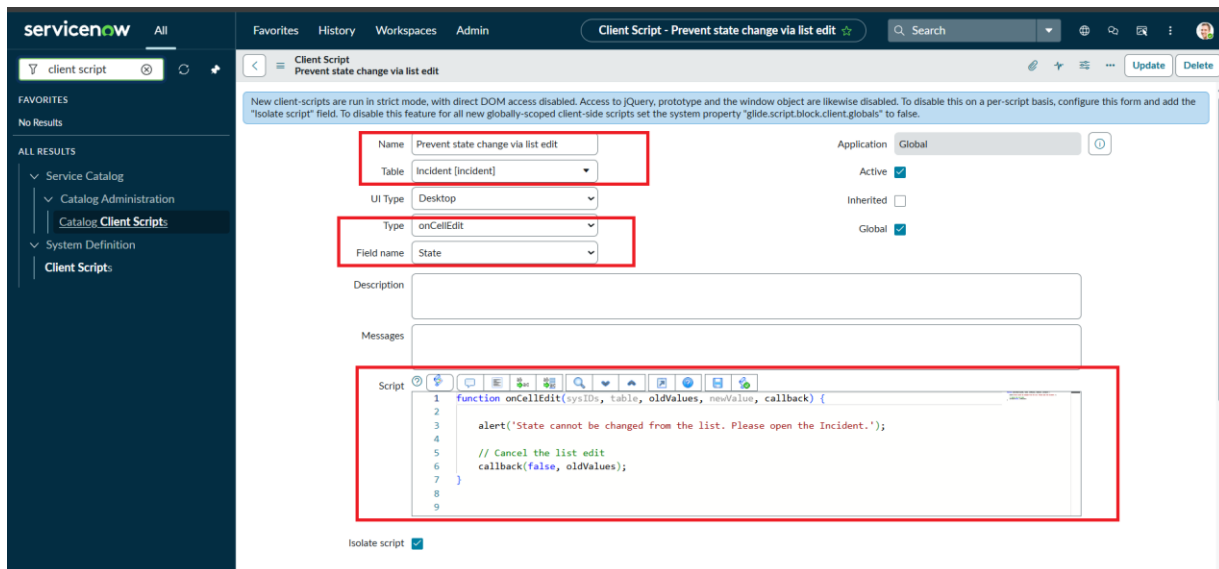
Screenshot 7: Saving the Urgency UI Policy Action.

5. Verify by opening an Incident, setting Impact to High, and confirming that Urgency becomes read-only.

Milestone 3: Create onChange Client Script

Activity 1: Create Client Script

1. Navigate to System UI → Client Scripts.
2. Click New to create a new Client Script.



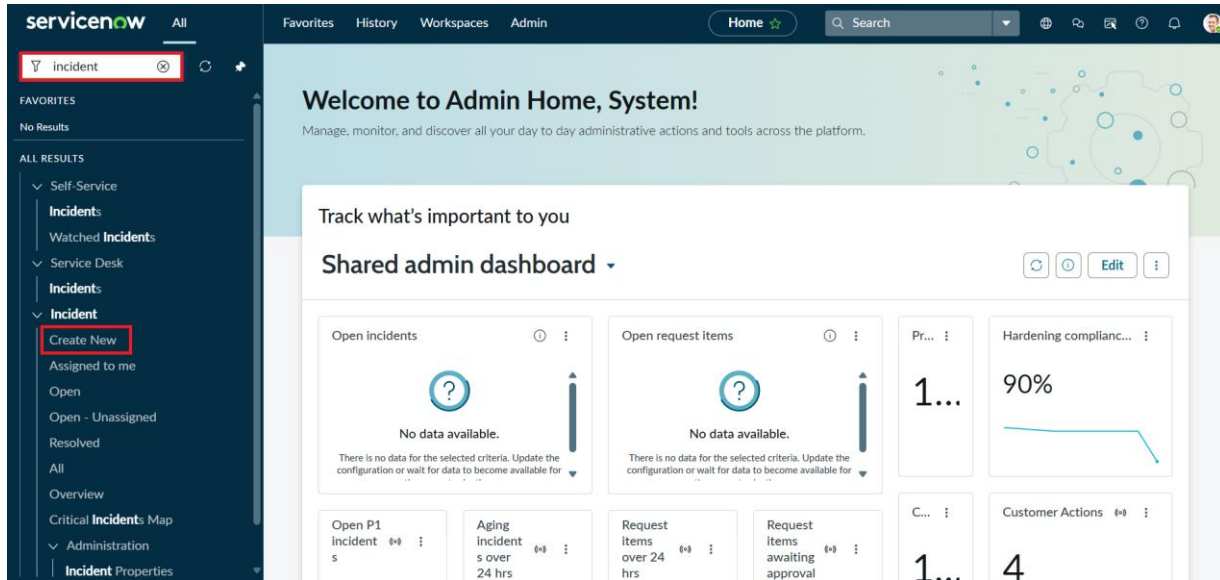
Screenshot 8: Creating the onChange Client Script.

Fill in the details:

- **Name:** Auto set urgency for high impact
- **Table:** Incident

- **Type:** onChange
- **Field name:** Impact
- **Active:** true

3. Paste the following script into the Script field:



Screenshot 9: Client Script editor and script configuration.

4. Click Submit to save the Client Script.

5. Verify by opening an Incident and changing Impact to High. The Urgency field should update automatically.

Milestone 4: Create onSubmit Client Script – Save Validation

Activity 1: Create Client Script

1. Navigate to System UI → Client Scripts and click New.
2. Fill in Name = Prevent save if Assigned To missing, Table = Incident, Type = onSubmit, and Active = true.

servicenow All

Incident - Create INC0010012

Incident New record

Number INC0010012

* Caller Adela Cervantsz

Category Inquiry / Help

Subcategory -- None --

Service

Service offering

Configuration item

Channel -- None --

State New

Impact 1 - High

Urgency 1 - High

Priority 1 - Critical

* Assignment group Software

Assigned to

* Short description testing UI policy

Description

Related Search Results

Related Search Q testing UI policy Knowledge & Catalog (AI)

Company policies No description

Whitelist IP Need to whitelist an IP address?

Submit Resolve

Screenshot 10: Configuring the onSubmit Client Script.

3. Paste the following script into the Script field:

servicenow All

Incident - Create INC0010012

Incident New record

Number INC0010012

* Caller Adela Cervantsz

Category Inquiry / Help

Subcategory -- None --

Service

Service offering

Configuration item

Channel -- None --

State New

Impact 1 - High

Urgency 1 - High

Priority 1 - Critical

* Assignment group Software

Assigned to Beth Anglin

* Short description testing UI policy

Description

Related Search Results

Related Search Q testing UI policy Knowledge & Catalog (AI)

Company policies No description

Whitelist IP Need to whitelist an IP address?

Order

Firewall Rule Change Cisco Firewall Appliance

Order

Screenshot 11: onSubmit Client Script configuration.

4. Click Submit to save the Client Script.

5. Verify by trying to save an Incident with Impact = High and Assigned To empty. The save should be prevented and a warning should appear.

Milestone 5: Create onCellEdit Client Script

Activity 1: Create Client Script

1. Navigate to System UI → Client Scripts and click New.

2. Fill in Name = Prevent state change via list edit, Table = Incident, Type = onCellEdit, Field name = State, and Active = true.

Screenshot 12: Configuring the onCellEdit Client Script.

3. Paste the following code into the Script field:

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:39:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF-TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Opspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2016-08-10	admin

Screenshot 13: onCellEdit Client Script.

4. Click Submit to save the Client Script.

5. Verify by opening an Incident list and trying to change the State field directly in the list.

Milestone 6: Testing the Configuration

Activity 1: Test Mandatory Enforcement

1. Navigate to Incident → Create New to open a new Incident form.

Incident - INC0008111

Number: INC0008111

* Caller: System Administrator

Category: Inquiry / Help

Subcategory: -- None --

Service: -- None --

Configuration Item: -- None --

* Short description: ATF: Test1

Description:

Country: -- None --

States: -- None --

Channel: -- None --

State: In Progress

Impact: 3 - Low

Urgency: 3 - Low

Priority: 5 - Planning

Assignment group: -- None --

Assigned to: -- None --

Save

- Add to Visual Task Board
- Copy Incident
- Create Incident Task
- Create Outage
- Create Problem
- Create Request
- Create Child Incident
- Create Normal Change
- Create Standard Change
- Create Emergency Change
- Refresh Impacted Services
- Metrics Timeline
- Follow on Live Feed
- Analyze Access
- Configure
- Export
- View
- Create Favorite
- Copy URL
- Copy sys_id
- Show XML
- History
- Reload form

Screenshot 14: Incident form used for testing.

2. Set Impact to High and leave Assigned To empty.
3. Click Submit and verify that the Incident is not saved because Assigned To is required.

dev354432.service-now.com says
State cannot be changed from the list. Please open the Incident.

Number	Opened	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Inquiry / Help	Software	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Inquiry / Help	Software	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Inquiry / Help	Software	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Inquiry / Help	Software	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	Inquiry / Help	Software	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	Inquiry / Help	Software	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Inquiry / Help	Software	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Inquiry / Help	Software	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Inquiry / Help	Software	(empty)	2020-06-07 09:04:34	admin
INC0001990	2016-08-10	Inquiry / Help	Software	(empty)	2016-08-10	admin

Screenshot 15: Validation message shown when Assigned To is missing.

4. Confirm that the UI Policy and Client Script behaviors are working as configured.

Activity 2: Test Successful Save

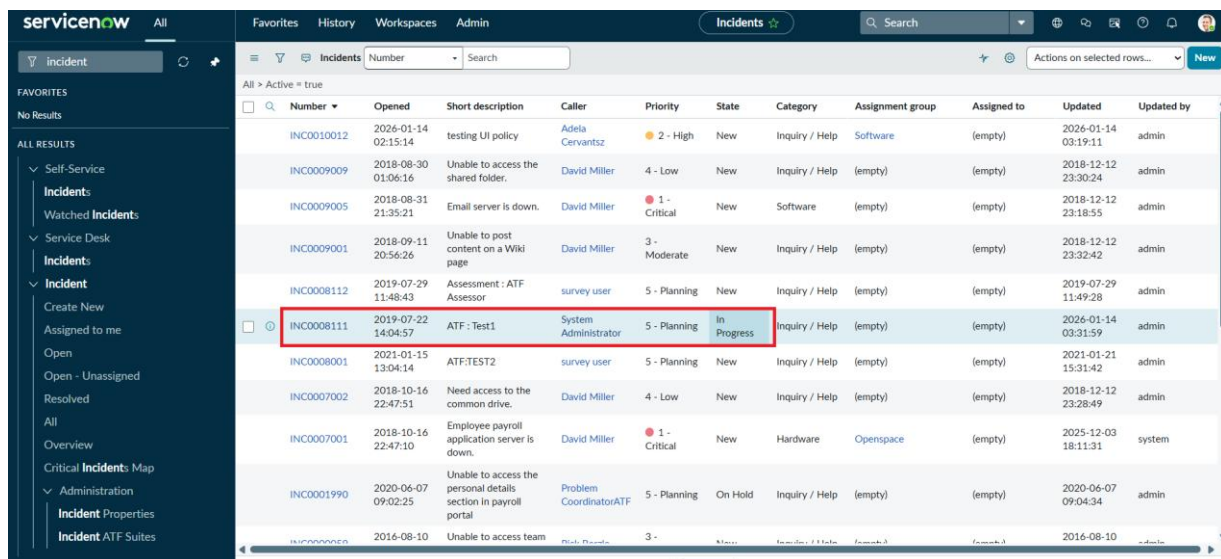
1. Open the Incident form and select a user in the Assigned To field.
2. Click Submit and verify that the Incident is saved successfully.
3. Confirm Urgency auto-setting, read-only behavior, and State restrictions.

Activity 3: Reverse Condition Test

1. Open an existing Incident where Impact is High.
2. Change Impact from High to Medium.
3. Verify that Assigned To is no longer mandatory and Urgency becomes editable.
4. Click Submit to save the record.

Activity 4: Test List Edit Blocking

1. Navigate to Incident → All to view Incident records.
2. Double-click the State field and attempt to edit it directly from the list.
3. Verify that the alert appears and the State value remains unchanged.



	Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
☐	INC0010012	2024-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2024-01-14 03:19:11	admin
☐	INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
☐	INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
☐	INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
☐	INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
☐	INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2024-01-14 03:31:59	admin
☐	INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
☐	INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
☐	INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
☐	INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
☐	INC0001990	2016-08-10	Unable to access team	David Miller	3 -	New	Inquiry / Help	(empty)	(empty)	2016-08-10	admin

Screenshot 16: Incident list used to verify list-edit behavior.

Activity 5: Test Form-Based Update

1. Open the same Incident record.
2. Change the State field on the Incident form.
3. Click Update.
4. Verify that the State change is saved successfully and other UI Policy/Client Script rules remain active.

Outcome

The configuration demonstrates the combined use of UI Policies and Client Scripts on the Incident form. The High Impact Control UI Policy makes Assignment group mandatory for high-impact incidents and controls the Urgency field. The onChange Client Script automatically sets Urgency to High, the onSubmit Client Script validates Assigned To before saving, and the onCellEdit Client Script blocks direct State changes through list editing.

Advantages

- Improves data consistency on Incident records.
- Enforces mandatory fields based on business conditions.
- Automatically updates field values when Impact changes.
- Prevents invalid or incomplete Incident submission.
- Controls list editing of sensitive fields such as State.
- Provides dynamic form behavior without requiring a full page submission.

Disadvantages

- Incorrect Client Script conditions can affect Incident form behavior.
- Multiple UI Policies and Client Scripts can become difficult to maintain.
- Changes in business requirements may require script updates.
- Client-side controls should be tested carefully to avoid unexpected user experience issues.

Future Enhancement

The project can be enhanced by adding additional validation rules, role-based field controls, reusable Script Includes, automated testing with ServiceNow ATF, and more detailed user guidance messages. Additional UI Policies can also be introduced for different Impact, Urgency, and Assignment conditions as business requirements evolve.

Conclusion

The Implement Client Script & UI Policy (Incident) micro project demonstrates how ServiceNow client-side controls can enforce dynamic field behavior, automate updates, and prevent incorrect data submission. By using UI Policy Actions together with onChange, onSubmit, and onCellEdit Client Scripts, the project helps maintain complete and consistent Incident information while improving form usability and data integrity.